

and download the entire course content anytime, anywhere. Also, these can deliver any format of file (html, pdf, reader, etc). So the students can use the materials actively. They would be able to make the materials their own while studying and this may help them organise the materials cognitively to support learning and remembering. Students can also download content for offline purposes such as studying off-line, communication and discussion with fellow students, group assignments and communication with the instructor – including submission of assignments with correction and feedback.

Mobile interactive learning objects for m-learning can generally be structured in the same way as learning objects for e-learning. The obvious differences remain in the presentation and amount of information offered. Due to limited screen size, very little continuous text can be used. Instead, different kinds of media should be applied. In this design, we support text and multimedia output.

The mobile learning architecture consists of four layers: application layer, integration through Web services standards layer, delivery layer and physical layer.

Application layer. The application

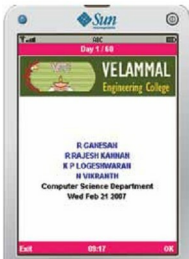


Fig. 2: User details

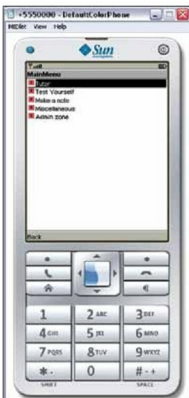


Fig. 3: Main menu

layer consists of various services for students and instructors. These services include teaching, group discussion, library, test and questionnaire, etc. They are created by tutors and administrators for students' use. The interaction between students, tutors and administrators is done at the application layer. The other layers below the application layer are completely transparent to students.

The integration through Web services standards layer. At this layer, the integration through Web services would integrate all the contents and applications that may already be available in different formats. The Web services architecture used for this purpose would cause the whole integration process to be similar to plug-and-play, and provide enough flexibility to allow content independent of devices. The architecture ensures availability, scalability and performance, as well as the ability

to simultaneously deliver data, voice and video. It also manages security, quality of service and content distribution.

The delivery layer (WWW). The delivery devices layer delivers the content using Internet-enabled multiple devices. The flexible services architecture supports all hand-held devices.

Some of the wireless technologies used in mobile learning are:

1. **Global System for Mobile Communications (GSM):** GSM is one of the leading digital cellular systems. It uses narrow-band time-division multiple access (TDMA). GSM provides integrated voice mail, high-speed data, fax, paging and short message services capabilities, as well as secure communications. It offers the best voice quality of any digital wireless standard at present.

2. **Wireless application protocol (WAP):** It is a free, unlicensed protocol for wireless communications. It makes possible creation of advanced communication services and access to Internet pages from a cellular phone. WAP is a *de-facto* industry standard supported by a large number of suppliers. WAP devices understand the WML language (an XML application) that is optimised for small screens and navigation without a keyboard. WAP also supports WMLScript scripting language.

3. **General packet radio service (GPRS):** It's a packet-linked technology that enables high-speed wireless Internet and other data communications. GPRS provides about four times greater speed than conventional GSM.

Physical layer. The physical layer consists of students, administrators and tutors. At this layer exists the direct interaction between the tutor and students, using handheld devices and telecast centre.

Prototype implementation

The m-learning technology environment may include mobile devices such as pocket PCs, mobile phones and portable keyboards. These m-learning devices have the power of a desktop that gives access to Microsoft Pocket application learning content. The communication components of a learning environment include resources (articles on the Web and references to other resource materials), online access to the discussion forum with the possibility of quick access for reading in the forum and writing contributions, and e-mail